

# OMID ZANGANEH

## Senior GIS Developer and Analyst

MS Geography — Geographic Information Science & Technology · Workflow Automation · AI/ML Integration

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### PROFESSIONAL SUMMARY

GIS Developer and Analyst specializing in workflow automation, ArcGIS Pro development, and AI-powered spatial tools. Builds production apps in Python and C#, ArcGIS Pro add-ins, geoprocessing toolboxes, and SQL Server pipelines that turn multi-day manual work into minutes. 2026 Edison Award Winner. MS Geography (GIS&T), 4.0 GPA.

### WORK EXPERIENCE

#### Olsson 2026 Edison Award Winner

March 2025 – Present

*GIS Developer and Analyst*

Lincoln, Nebraska

- Built production Python and C# GIS apps, including bore profile automation that cut processing from days to minutes.
- Created an ArcGIS Automation Suite of Python toolboxes that reduced manual GIS steps by ~90% and sped up fiber network design.
- Shipped ArcGIS Pro add-ins (.NET/C#/WPF): GIS Data Downloader, FTTH Network Designer, RF Analysis panel, and Street View map tool.
- Built AI tools with Azure OpenAI and Google AI Studio — including RFP Radar — cutting sourcing timelines from months to hours; plus YOLO apps for utility infrastructure inventory.

#### University of Nebraska at Omaha

January 2024 – August 2025

*Graduate Teaching Assistant — Instructor of Record*

Omaha, Nebraska

- Taught Human-Environment Geography labs as sole instructor of record to 150+ students across three semesters.

#### University of Nebraska at Omaha

June 2024 – August 2025

*GIS Technician — Omaha Spatial Justice Project*

Omaha, Nebraska

- Digitized historical parcels and legal records into a geodatabase of racially restrictive covenants in Douglas County.

### SELECTED PROJECTS

#### RFP Radar — AI-Powered RFP Sourcing

Azure OpenAI · Gemini · Playwright · Python/C#

Desktop app for web-grounded RFP search and classification. Months of sourcing → hours.

#### Bore Profile Automation — Directional Drilling Profiles

Python · C# · ArcGIS Pro · SQL Server

Reads waypoints and elevation models; generates 2D/3D bore profiles. Days → minutes.

#### Fiber Automatic Expansion — Fiber Build-Area Planning

C# · ArcGIS Pro SDK · WPF

Grids study areas, finds viable build zones by PPM thresholds, and scores economics. Days/weeks → minutes.

#### FTTH Network Designer — Fiber Optic Network Planning

C# · ArcGIS Pro SDK · Kruskal MST

Places shafts, connects homes, and builds an optimal trunk/branch backbone. Days → minutes.

#### ArcGIS Data Downloader — Multi-Source GIS Acquisition

C# · .NET 8 · ArcGIS Pro SDK · REST APIs

Downloads OSM, USGS, FEMA, Census/TIGER, and BSL layers into ArcGIS Pro projects.

#### Aerial & Streetview AI Detection — YOLO Utility Detection

Python · YOLO · OpenCV

Detects utility assets from aerial and street imagery; exports georeferenced results to ArcGIS.

## EDUCATION

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### Master of Science, Geography — Geographic Information Science & Technology August 2025

University of Nebraska at Omaha, Nebraska **GPA: 4.00** · **GRACA Project Award**

**Thesis:** Spatiotemporal Analysis of NOx Emissions from U.S. Cement Plants Using TROPOMI Data — remote sensing, hotspot analysis, and environmental justice.

### Bachelor of Science, Geomatics (Surveying) Engineering

August 2016

Geomatics College of National Cartographic Center (GCNCC), Tehran

## TECHNICAL SKILLS

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**Languages & Frameworks** Python · C# / .NET · SQL · TypeScript · JavaScript · React · Next.js · WPF

**GIS & Spatial** ArcGIS Pro · ArcGIS Pro SDK · ArcPy · Python Toolboxes · QGIS · Network Analysis · Remote Sensing · Google Earth Engine

**AI / ML** Azure OpenAI · Azure AI Foundry · Google Gemini · YOLO · OpenCV · Prompt Engineering

**Data & Backend** SQL Server · PostgreSQL · Supabase · ETL Pipelines · pyodbc

**Web & Cloud** Next.js · Leaflet · Vercel · REST APIs · Microsoft Azure

**Tools** CustomTkinter · PyInstaller · Playwright · AutoCAD · Tableau

**Languages** English (Fluent) · Persian (Native)

For project write-ups, live GIS tools, and demos: [omidzanganeh.com](https://omidzanganeh.com)

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